Measures

- Information(A) $= \psi(p)$
  ↳ $\psi(\cdot) \to$ ① ≥ 0, ② $= 0$ if $p = 1$ / ③ (monotonicity) $p \leq p' \to \psi(p) \geq \psi(p')$
  ④ (continuity) ⑤ (additive under $\perp$) $\psi(p_1 \cdot p_2) = \psi(p_1) + \psi(p_2)$, for $p_1 \perp p_2$
  ↳ log satisfies $[\log_2 \frac{1}{p}]$
- Shannon's property:
  ① continuity ② (uniform case): $i = \frac{1}{N} \forall i = N$, then $\psi(p)$ is increasing in N (large sets > smaller sets)
  ③ (successive decisions)
  $\psi(p_1, \dots, p_N) = \psi(p_1 + p_2, p_3, \dots, p_N) + (p_1 + p_2)\psi\left(\frac{p_1}{p_1+p_2}, \frac{p_2}{p_1+p_2}\right)$

- Entropy (Shannon), avg bits of info $[\log_2 \frac{1}{p}]$ learnt
  ↳ $H(X) = \mathbb{E}_{x \sim P_X}[\log_2 \frac{1}{P_X(x)}] = \sum_x P_X(x) \log_2 \frac{1}{P_X(x)}$
  ↳ where X is discrete RV w/ PMF P_X
  ↳ ① $\log \frac{1}{0} = 0$ → Imp → $0 \log_2 \frac{1}{p} = 0$
- Bernoulli(p): $H_2(p) = p \log_2 \frac{1}{p} + (1-p) \log_2 \frac{1}{1-p}$
  ↳ sym. around p
- Uniform source: $P_X(x) = \frac{1}{|\mathcal{X}|} \forall x \in \mathcal{X}$
  $\Rightarrow H(X) = \mathbb{E}[\log_2 \frac{1}{1/|\mathcal{X}|}] = \log_2 |\mathcal{X}|$
- Joint entropy: $H(X, Y) = \mathbb{E}_{(x,y) \sim P_{XY}}[\log_2 \frac{1}{P_{XY}(x,y)}]$
  $= \sum_{x,y} P_{XY}(x,y) \log_2 \frac{1}{P_{XY}(x,y)}$
- conditional/prob: $H(Y|X) = \mathbb{E}_{x \sim P_X, y \sim P_{Y|X}}[\log_2 \frac{1}{P_{Y|X}(y|x)}]$
  avg uncertainty in Y given X
  $= \sum_{x,y} P_{XY}(x,y) \log_2 \frac{1}{P_{Y|X}(y|x)}$
  $= \sum_x P_X(x) H(Y|X=x)$
  $= \sum_x \sum_y P_{Y|X}(y|x) \log_2 \frac{1}{P_{Y|X}(y|x)}$

- Properties of Entropy
  ① $H(X) \geq 0$ (by log) ② $H(X) \leq \log_2 |\mathcal{X}|$ ($\overline{w}$ == → uniform)
  ③ chain rule: $H(X, Y) = H(X) + H(Y|X) = H(Y) + H(X|Y)$
  ④ chain rule (general): $H(X_1, \dots, X_n) = \sum_{i=1}^n H(X_i | X_1, \dots, X_{i-1})$
  ⑤ conditional reduces entropy: $H(X|Y) \leq H(X)$
  ↳ $\overline{w}$ == iff $X \perp Y$
  ⑥ sub-additivity: $H(X_1, \dots, X_n) \leq \sum_{i=1}^n H(X_i)$
  ↳ $\overline{w}$ == iff $X_1, \dots, X_n$ are $\perp$
  ↳ + $H(X|Y) \leq \log_2 |\mathcal{X}|$

KL divergence (relative entropy)

- $D(P \| Q) = \sum_x P(x) \log_2 \frac{P(x)}{Q(x)}$
  $= \mathbb{E}_{x \sim P}[\log_2 \frac{P(x)}{Q(x)}]$
  ↳ $D(P \| Q) \geq 0$, $\overline{w}$ == iff $P = Q$

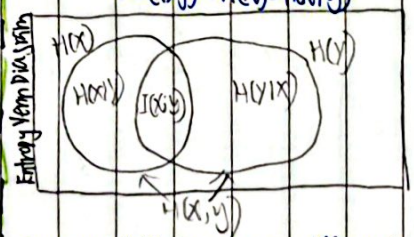
Mutual info (between RV)

- $I(X;Y) = H(Y) - H(Y|X)$
- [Joint] $I(X_1, X_2; Y_1, Y_2) = H(Y_1, Y_2) - H(Y_1, Y_2 | X_1, X_2)$
- [cond.] $I(X;Y|Z) = H(Y|Z) - H(Y|X, Z)$

(Notes) → $X \perp Y \to H(Y|X) = H(Y) \to I(X;Y) = 0$
→ $Y = X \to H(Y|X) = H(X|X) = 0 \to I(X;Y) = H(X)$
↳ when $p = \frac{1}{2}$, $I(X;Y) = 1 - H_2(\delta)$, for $H(Y|X) = H_2(\delta)$ & $H(Y) = H_2(p(1-\delta) + (1-p)\delta)$
[channel capacity]

Properties of MI

- [alternative form] $I(X;Y) = D(P_{XY} \| P_X \times P_Y)$
  $= \mathbb{E}[\log_2 \frac{P_{XY}(X,Y)}{P_X(X) P_Y(Y)}] = \sum_{x,y} P_{XY} \log_2 \frac{P_{XY}}{P_X P_Y}$
  $= \mathbb{E}[\log_2 \frac{P_{Y|X}}{P_Y}] = \sum_{x,y} P_{XY} \log_2 \frac{P_{Y|X}}{P_Y}$
- [symmetry] $I(X;Y) = H(X) + H(Y) - H(X,Y)$
  → $I(X;Y) = I(Y;X)$
  → $I(X;Y) = H(X) - H(X|Y)$



- [Non-negativity] $I(X;Y) \geq 0$ $\overline{w}$ == iff $X \perp Y$
- [upper bounds] $I(X;Y) \leq H(X) \leq \log_2 |\mathcal{X}|$
  $I(X;Y) \leq H(Y) \leq \log_2 |\mathcal{Y}|$
- [chain rule] $I(X_1, \dots, X_n; Y) = \sum_{i=1}^n I(X_i; Y | X_1, \dots, X_{i-1})$
- [data processing inequality] $X \to Y \to Z$, $X \perp Z | Y$
  → $I(X;Z) \leq I(X;Y)$ | $P_{Z|XY} = P_{Z|Y}$, $P_{X|YZ} = P_{X|Y}$
  → $I(X;Z) \leq I(Y;Z)$
- [partial sub-additivity]
  If $(X_1, \dots, X_n) \perp (X_1, \dots, X_n)$ + Y_i depends only thru X_i → $I(X_1, \dots, X_n; Y_1, \dots, Y_n) \leq \sum_{i=1}^n I(X_i; Y_i)$
  → $P(Y_1, \dots, Y_n | X_1, \dots, X_n) = P(Y_1 | X_1) \cdots P(Y_n | X_n)$

Symbol Source

- map symbol $x \in \mathcal{X}$ (source) to bin seq $C(x)$ w/ $\ell(x)$ length
- $L(C) = \sum_{x \in \mathcal{X}} P_X(x) \ell(x)$ ↳ avg length of (C.)
- Decodability condition
  ↳ nonsingular: $C(x) \neq C(x') \forall x \neq x'$
  ↳ [uniquely decodable] $x_1, \dots, x_n$ can be uniquely identified from $C(x_1) \cdots C(x_n)$
  ↳ $C(x_1) \cdots C(x_i) \,!= C(x_1') \cdots C(x_i')$
  ↳ [prefix-free] ☆ no codeword is a prefix of other ↳ $C(x) + b_1 \cdots b_i = C(x')$ ×
  ↳ easy/efficient to decode
  aka. instantaneous code

Kraft Inequality (how small can L(C) be)

- Theorem: $\forall$ prefix-free code (C.): $x \in \mathcal{X}$
  → (C.) of $\ell(x)$: $\sum_{x \in \mathcal{X}} 2^{-\ell(x)} \leq 1$ (necessary)
  (proof by tree) - prob
- Theorem (Existence Property) If $\{\ell(x)\}_{x \in \mathcal{X}}$ satisfies $\sum_{x \in \mathcal{X}} 2^{-\ell(x)} \leq 1$ → can construct prefix-free code $x \to C(x)$ of $\ell(x)$ (sufficiency) $\forall$

Entropy bound (fundamental limit)

- (Theorem.) $X \sim P_X$ & PFC (C.): $L(C) \geq H(X)$
  ↳ $\overline{w}$ == iff $P(x) = 2^{-\ell(x)} \forall x \in \mathcal{X}$
  ↳ proof idea: $L(C) - H(X) \geq 0$ [KL]
- Implication: powers of 2 converges → $H(X)$

Shannon-Fano Code

$\ell(x) = \lceil \log_2 \frac{1}{P_X(x)} \rceil$

- (Theorem): $H(X) \leq L(C) < H(X) + 1$
  ↳ less sig if $H(X)$ is large
- (Mismatched case): True dist: $P(x)$, chosen: $Q(x)$
  ↳ $H(X) + D(P_X \| Q_X) \leq L(C) < H(X) + D(P_X \| Q_X) + 1$

Huffman code

- (Theorem): no uniquely decodable symbol source can achieve smaller $L(C)$ than Huffman code
  → also $H(X) \leq L(C) < H(X) + 1$

- Key limitation to symbol code: memoryless
- symbol grp: $\mathcal{X}^k$, $P_{X_1, \dots, X_k}$ + even $\overline{w}$ ⊥ symbols: $NH(X) \leq L(C) \leq NH(X) + 1$
  ↳ $H(X) \leq \frac{L(C)}{N} \leq H(X) + \frac{1}{N}$
  ↳ but diff. to find $P_{X_1, \dots, X_N}$, sorting $|\mathcal{X}|^N$ prob

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Useful Prob properties

- $\sum_x P_X(x) = 1$
- $\mathbb{E}[cX] = c\mathbb{E}[X]$ ○ $(c \circ E)$ $\mathbb{E}[X+Y] = \mathbb{E}[X] + \mathbb{E}[Y]$
- $X \perp Y \to \mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$ ○ $\mathbb{P}[\mathbb{E}[A]] = \mathbb{P}(A)$
- (LOTP) $\mathbb{P}[A] = \sum_x P_X(x) \mathbb{P}[A | X = x]$
- (LOTE) $\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y|X]]$ ○ $P_{XY} = P_X P_Y$ if $X \perp Y$
- (i) $P_{X|YZ} = P_{X|Z}$ $P_{Y|XZ} = P_{Y|Z}$ ≡ (ii) $P_{X|YZ} = P_{X|Z}$ $\forall x, y, z$, $X \perp Y$ given Z
- $X \perp Y$, then $f(X) \perp g(Y)$, f, g is deterministic
- $\perp$ joint: $P_{X_1 \dots X_n} = \prod_{i=1}^N P_{X_i}$ (pairwise ⊥ ↛ joint)
- $\mathrm{Var}[X] = \mathbb{E}[(X - \mu)^2] = \mathbb{E}[X^2] - \mu^2 = \sigma^2$
- $\mathrm{Var}[cX] = c^2 \mathrm{Var}[X]$ ○ $X \perp Y \to \mathrm{Var}[X+Y] = \mathrm{Var}[X] + \mathrm{Var}[Y]$
- $\mathrm{Cov}[X, Y] = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)]$, $\mu_i := \mathbb{E}[i]$
- $P_X(x) = \sum_y P_{XY}(x,y)$, $P_Y(y) = \sum_x P_{XY}(x,y)$
- $\mathbb{P}[A \cup B] = \mathbb{P}[A] + \mathbb{P}[B] - \mathbb{P}[A \cap B]$
  (union bound) $\mathbb{P}[\bigcup_{i=1}^N A_i] \leq \sum_{i=1}^N \mathbb{P}[A_i]$
- (law of ↑#) $X_1, \dots, X_N$ iid w/ μ → $\mathbb{P}[|\frac{1}{N}\sum_{i=1}^N X_i - \mu| > \epsilon] \to 0$ as $n \to \infty$ for arbitrary small $\epsilon > 0$
- $\log xy = \log x + \log y$ ○ $\log \frac{1}{x} = -\log x$
- $\log \frac{x}{y} = \log x - \log y$ ○ $\log x^c = c \log x$
- $\log_a x = \frac{\log_b x}{\log_b a}$ ○ $\log_e x \leq x - 1$ $\overline{w}$ == iff $x = 1$
- $\frac{d}{dx} x^c = c x^{c-1}$ ○ $\frac{d}{dx} e^{cx} = c e^{cx}$ ○ $\frac{d}{dx} \ln x = \frac{1}{x}$
- (chain rule) $\frac{d}{dx} f(g(x)) = f'(g(x)) g'(x)$
- (product rule) $\frac{d}{dx}(f(x) g(x)) = f'(x) g(x) + f(x) g'(x)$
- (quotient) $\frac{d}{dx}(f(x)/g(x)) = \frac{f'(x) g(x) - f(x) g'(x)}{g(x)^2}$
- $a \leq \lceil a \rceil < a + 1$ ○ $A < B \to \frac{1}{A} > \frac{1}{B}$

- Note: prefix-free → UD | UD ↛ prefix-free
- Note: prefix-free → kraft, kraft ↛ prefix-free
- WEAK LLN: $\lim_{n \to \infty} \mathbb{P}[|\frac{1}{n}\sum_{i=1}^n X_i - \mathbb{E}[X]| > \epsilon] = 0$
  (equivalently w/ high prob:
  $\lim_{n \to \infty} \frac{1}{n}\sum_{i=1}^n X_i = \mathbb{E}[X]$
- Σ of cont & discrete RV: $Y = X + Z$ | $X \in \{0, 1\}$, $Z \sim \mathrm{Poi}$
  ↳ $F_Y(y) = \mathbb{P}(Y \leq y) \cdots = \sum_x F_Z(y-x) P_X(x)$
  ↳ $P_Y(y) = \frac{d}{dy} F_Y(y) \cdots = \sum_x P_Z(y-x) \cdot P_X(x)$

Block source $X^n \to \{1 \dots M\} \to \hat{X}^n$

- $R = \frac{1}{n} \log_2 M$ (bits/symbol) ↑ better +P_e ↓
- fixed length SC theorem (trade off): $\forall$ discrete memoryless source w/ per-symbol dist P_X:
  - (Achievability) $R > H(X)$: $\forall \epsilon > 0$, $\exists n$ & SC of R: $P_e \leq \epsilon$
  - (Converse) $R < H(X)$: $\forall \epsilon > 0$, $\forall$ SC of R, irrespective of n, $P_e > \epsilon$

(Asymptotic equipartition) Typical set & AEP

- $X = (X_1, \dots, X_n) :=$ iid seq with each $X_i \sim P_X$; $P_X(x) = \prod_{i=1}^n P_X(x_i)$ dist of X
- $T_n(\epsilon) = \{x \in \mathcal{X}^n : 2^{-n(H(X)+\epsilon)} \leq P_X(x) \leq 2^{-n(H(X)-\epsilon)}\}$
  → ϵ is fixed constant
- Properties:
  1. $H(X) - \epsilon \leq \frac{1}{n} \sum_{i=1}^n \log_2 \frac{1}{P_X(x_i)} \leq H(X) + \epsilon$
  2. $\mathbb{P}[X \in T_n(\epsilon)] \to 1$ as $n \to \infty$
  3. $|T_n(\epsilon)| \leq 2^{n(H(X)+\epsilon)}$
  4. $|T_n(\epsilon)| \geq (1 - o(1)) 2^{n(H(X)-\epsilon)}$, $o(1) \to 0$ as $n \to \infty$
- map typical seq → $\{1, \dots, M-1\}$ & non-typical → M
  ↳ $\hat{X}$ arbitrary, if $m = M$
- possible for $M = |T_n(\epsilon)| + 1$
  → $P_e \leq \mathbb{P}[X \notin T_n(\epsilon)]$
- $M = 2^{n(H(X)+\epsilon)} + 1$
  ↳ note: only holds ($P_e \downarrow$) when $n \to \infty$; design $R := H(X)$ (smaller ϵ) & smaller P_e ϵ → higher n

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(Converse) Fano's inequality

- Y discrete RV X & $\hat{X}$ on common finite $\mathcal{X}$,
  $H(X|\hat{X}) \leq H_2(P_e) + P_e \log_2(|\mathcal{X}| - 1)$
  ↳ Proof. $H(X|\hat{X}) \overset{(a)}{=} H(X, E|\hat{X})$
  $\overset{(b)}{=} H(E|\hat{X}) + H(X|\hat{X}, E)$
  $\overset{(c)}{\leq} H(E) + H(X|\hat{X}, E)$
  $\overset{(d)}{=} H_2(P_e) + P_e H(X|\hat{X}, E=1) + (1 - P_e) H(X|\hat{X}, E=0)$
  $\overset{(e)}{\leq} H_2(P_e) + P_e \log_2(|\mathcal{X}| - 1)$
  → (a): E is deterministic func of $(X, \hat{X})$
  ↳ $H(X, E|\hat{X}) = H(X|\hat{X}) + H(E|X, \hat{X})$ → $= 0$
  → (b) chain rule | (c) cond ↓ entropy
  → (d) $H(E) = H_2(P_e)$ & defn. of $H(\cdot)$ 2nd term
  ↳ (e) since X has no uncertainty given $\hat{X}$ when $E = 0$ + $\frac{1}{|\mathcal{X}|}$ is due to X taking $|\mathcal{X}| - 1$ values given $\hat{X}$ when $E = 1$, max $H(\cdot)$

converse proof - cont.

- Theorem: $R < H(X) \to P_e = P[\hat{X} \neq X] \not\to 0$ as $n \to \infty$

↳ Proof: 1. For convenience, $\log_2(|\mathcal{X}^n| - 1) \le \log_2 |\mathcal{X}^n|$
$= n \log_2 |\mathcal{X}|$

2. $H(X|\hat{X}) \le P_e \cdot n \log_2|\mathcal{X}| + 1$ ($H_2(P_e) \le 1$)

3. $\therefore I(X;\hat{X}) \ge H(X) - P_e \cdot n \log_2|\mathcal{X}| - 1$

Be def. of $H(X)$ + discrete ⊥ $= nH(X) - P_e \cdot n \log_2|\mathcal{X}| - 1$

4. alternative def. of $I(\cdot)$
$\to I(X;\hat{X}) = H(\hat{X}) - H(\hat{X}|X)$
$\le H(\hat{X})$ (non-negativity of $H(\cdot)$)
$\le nR$ ($\hat{X}$ takes $M = 2^{nR}$ values & max entropy bound)

5. $\therefore nR \ge nH(X) - P_e \cdot n \log_2|\mathcal{X}| - 1$
$\to P_e \ge \frac{1}{\log_2|\mathcal{X}|}\left[H(X) - R - \frac{1}{n}\right]$

6. Clearly, $R < H(X) \to P_e \not\to 0$ as $n \to \infty$

↳ this sc def. for arbitrary n, not only $n \to \infty$, result for $n \to \infty$ implies arbitrary n
∵ only way to set arbitrarily small error prob at fixed n → $P_e = 0$
↳ if $P_e = 0$ at n (finite), then also can for $n \to \infty$ simply by using code many times in succession

Channel

source X → encoder → m → encoder → codeword $\tilde{X}$ → channel → $\tilde{Y}$ → decoder → $\hat{m}$ → source decoder → $\hat{X}$

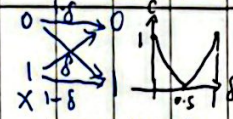
channel: $\tilde{X}$ → channel → $\tilde{Y}$, $P_{Y|X}(y|x)$

- $x \in \{0,1\}^k$, $y \in \{0,1\}^k$, $P_{Y|X} = \prod_{i=1}^{k} P_{Y|X}(y_i|x_i)$
- $x^{(m)} \in \{0,1\}^n$ transmitted over n uses of channel
- $P_e = P[\hat{m} \neq m]$ → consider avg over M msg
- goal: high M, rate $R = \frac{1}{n}\log_2 M$ bits/channel use
  ↳ #msg: $M = 2^{nR}$, n is block length
- Channel cap: $\max R$ s.t. $\forall \epsilon > 0$ ~~$\exists n$~~
  $\exists n$ & $C = \{x^{(1)}, \dots, x^{(M)}\}$, $M = 2^{nR}$, $P_e \le \epsilon$
  ↳ Channel coding Theorem: $C = \max_{P_X} I(X;Y)$ for discrete memoryless channel $P_{Y|X}$
  ↳ Proof: (Achievable) $\forall R < C, \exists R(\tilde{n}) \ge R$ w ↓ P_e
  (converse): $\forall R > C, \forall n \ge R$ w ↓ P_e

- Noiseless channel: $H(X|Y) = 0$ ($X = Y$)
  $I(X;Y) = H(X) - H(X|Y) = H(X)$
  $C = \max_{P_X} I(X;Y) = \max_{P_X} H(X) = 1$

More types of channel

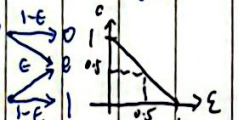
- BSC: $X = Y = \{0,1\}$:
  $P_{Y|X}(y|x) = \begin{cases} 1-\delta & y = x \\ \delta & y = 1-x \end{cases}$



$H(Y|X) = \sum_x P_X(x) H(Y|X=x)$
$= P H(Y|X=1) + (1-P) H(Y|X=0)$
$= P H_2(\delta) + (1-P) H_2(\delta)$
$= H_2(\delta)$

- notice $P_Y = P_X$ (with flipped) when $P_0 = \frac{1}{2}$
  $P_Y(y) = P(Y=y|X=0) + P(Y=y|X=1)$
  ↳ $P(Y=1) = \delta + P - 2P\delta$, $P(Y=0) = 1 - \delta - P + 2P\delta$ → $H(Y) = H_2(\delta + P - 2P\delta)$
- $C = \max_{P_X} I(X;Y)$
  $= \max_{P_X} (H(Y) - H_2(\delta))$
  $= 1 - H_2(\delta)$ (P is $\frac{1}{2}, \frac{1}{2}$ uniform)

- BEC: $X = \{0,1\}$, $Y = \{0,1,e\}$:
  $P_{Y|X}(y|x) = \begin{cases} 1-\epsilon & y = x \\ \epsilon & y = e \\ 0 & y = 1-x \end{cases}$



- $I(X;Y) = H(X) - H(X|Y)$
  1. $H(X|Y) = \sum_y P_Y(y) H(X|Y=y)$
     $= P_Y(e) H(X|Y=e)$ [$H(\cdot) = 0$ for $P = 0$ or 1]
     $= [P(Y=e|X=1) + P(Y=e|X=0)] H(X|Y=e)$ (chain rule)
     $= \epsilon H(X|Y=e) = \epsilon H(X)$
  2. ∵ $P_{X|Y}(0|e) = \frac{P_X(0)\epsilon}{P_X(0)\epsilon} = P_X(0)$
     & $P_{X|Y}(1|e) = \frac{P_X(1)\epsilon}{P_X(1)\epsilon} = P_X(1)$
     → $H(X|Y=e) = H(X)$
  3. ∴ $I(X;Y) = (1-\epsilon) H(X)$
  4. $C = 1 - \epsilon$, $H(X) = 1$

Jointly typical seq properties

$P_{X,Y}(x,y)$ conditions: $x \in \mathcal{X}, y \in \mathcal{Y}$, $T_n(\epsilon)$

- $2^{-n(H(X)+\epsilon)} \le P_X(x) \le 2^{-n(H(X)-\epsilon)}$
- $2^{-n(H(Y)+\epsilon)} \le P_Y(y) \le 2^{-n(H(Y)-\epsilon)}$
- $2^{-n(H(X,Y)+\epsilon)} \le P_{X,Y}(x,y) \le 2^{-n(H(X,Y)-\epsilon)}$

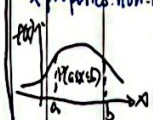
Jointly typical seq properties

1. (Equivalent def): $(x,y) \in T_n(\epsilon)$ iff:
   - $H(X) - \epsilon \le \frac{1}{n}\sum_{i=1}^{n} \log_2 \frac{1}{P_X(x_i)} \le H(X) + \epsilon$
   - $H(Y) - \epsilon \le \frac{1}{n}\sum_{i=1}^{n} \log_2 \frac{1}{P_Y(y_i)} \le H(Y) + \epsilon$
   - $H(X,Y) - \epsilon \le \frac{1}{n}\sum_{i=1}^{n} \log_2 \frac{1}{P_{X,Y}(x_i,y_i)} \le H(X,Y) + \epsilon$
2. (High Prob): $P[(X,Y) \in T_n(\epsilon)] \to 1$ as $n \to \infty$
3. $|T_n(\epsilon)| \le 2^{n(H(X,Y)+\epsilon)}$
4. If $(X',Y') \sim P_X(x')P_Y(y')$ are ⊥ copies of (X,Y), then:
   $P[(X',Y') \in T_n(\epsilon)] \le 2^{-n(I(X;Y)-3\epsilon)}$
   ↳ proof: $P[(X',Y') \in T_n(\epsilon)] = \sum_{(x,y)\in T_n(\epsilon)} P_X(x')P_Y(y')$
   ∵ $P_X(x') \le 2^{-n(H(X)-\epsilon)}$ & $P_Y(y') \le 2^{-n(H(Y)-\epsilon)}$ ← $\le \sum_{(x,y)\in T_n(\epsilon)} 2^{-n(H(X)-\epsilon)} 2^{-n(H(Y)-\epsilon)}$
   ∵ property 3 ← $\le |T_n(\epsilon)| 2^{-n(H(X)-\epsilon)} 2^{-n(H(Y)-\epsilon)}$
   ∴ $I(X;Y) = H(X) + H(Y) - H(X,Y)$ ← $= 2^{-n(I(X;Y)-3\epsilon)}$

Continuous

PDF f_X: $h(X) = \mathbb{E}_{f_X}[\log_2 \frac{1}{f_X(X)}] = \int_{\mathcal{X}} f_X(x) \log_2(\cdot) dx$

- $h(X,Y) = \mathbb{E}_{f_{X,Y}}[\log_2 \frac{1}{f_{X,Y}(x,y)}]$
- $f_Y(y) = \int f_X(x) f_{Y|X}(y|x)$
- properties: chain rule: $h(X_1, \dots, X_n) = \sum_{i=1}^{n} h(X_i | X_1, \dots, X_{i-1})$
  ✓ cond ↓ entropy: $h(X|Y) \le h(X)$ (add-cond: $h(X_1, \dots, X_n) \le \sum_{i=1}^{n} h(X_i)$)
  ✓ $h(X) = h(X+c)$ for constant c
- X properties: non-negativity, invariance under one-one transform.
  ↳ if $Y = cX$, $f_Y(y) = \frac{1}{|c|} f_X(\frac{y}{c})$
  $h(Y) = h(X) + \log|c|$, ↳ $c \to 0$, $\log(c) \to -\infty$



- Uniform: $f_X(x) = \frac{1}{b-a}$ for $a < x < b$ → $h(X) = \log_2(b-a)$
  = 0 elsewhere
- uni Gaussian: $f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/(2\sigma^2)}$ → $h(X) = \frac{1}{2}\log_2(2\pi e \sigma^2)$, $X \sim N(\mu, \sigma^2)$
- KL: $D(f\|g) = \int_{\mathbb{R}} f(x) \log_2 \frac{f(x)}{g(x)} dx = \mathbb{E}_{\sim f}[\log_2 \frac{f_X}{g_X}]$
- MI: $I(X;Y) = D(f_{X,Y} \| f_X \times f_Y) = \mathbb{E}_{f_{X,Y}}[\log_2 \frac{f_{X,Y}}{f_X f_Y}]$
  $= h(Y) - h(Y|X) = h(X) - h(X|Y)$ — all KL properties

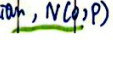
- MAX Entropy (univariate): $h(X) \le \frac{1}{2}\log_2(2\pi e \, Var[X])$, $\tilde{w}$ iff X is Gaussian (RV $X \sim \tilde{w}$ pdf f_X over $Var[X]$)

Gaussian Channel

PDF $f_{Y|X}$ | additive: $Y = X + Z$

↳ $f_{Y|X}(y|x) = f_Z(y-x)$ | $Z \sim N(0, \sigma^2)$: AWGN, $\sigma^2 > 0$

- (Power constraint) $\mathbb{E}[X^2] \le P$, $C = \{x^{(1)}, \dots, x^{(M)}\}$, $x^{(m)} = (x_1^{(m)}, \dots, x_n^{(m)})$
  ① $\frac{1}{n}\sum_{i=1}^{n}(x_i^{(m)})^2 \le P$, $\forall m \in [1, M]$
  or ② (relaxed) $\frac{1}{M}\sum_{m=1}^{M}\frac{1}{n}\sum_{i=1}^{n}(x_i^{(m)})^2 \le P$ → same capacity
- $C(P) = \max_{f_X: \mathbb{E}_{f_X}[X^2] \le P} I(X;Y)$
  ↳ AWGN: $C(P) = \frac{1}{2}\log_2(1 + \frac{P}{\sigma^2})$
  + cap-achieving f_X is Gaussian, $N(0, P)$



Gaussian Channel - cont

- Proof: fixed f_X s.t. $\mathbb{E}[X^2] \le P$
  1. $I(X;Y) = h(Y) - h(Y|X)$ (By def)
     $= h(Y) - h(X+Z|X)$ ($Y = X+Z$)
     $= h(Y) - h(Z|X)$ (X fixed, cond. on X)
     $= h(Y) - h(Z)$ ($X \perp Z$)
  2. Z is Gaussian → $h(Z) = \frac{1}{2}\log_2(2\pi e \sigma^2)$
  3. $Var[Y] = Var[X] + Var[Z]$ ($X \perp Z$ & $Y = X+Z$)
     $\le P + \sigma^2$ ($Var[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 \le \mathbb{E}[X^2] \le P$)
  4. Max Ent → $h(Y) \le \frac{1}{2}\log_2(2\pi e(P+\sigma^2))$
  5. ∴ $I(X;Y) \le \frac{1}{2}\log_2(2\pi e(P+\sigma^2)) - \frac{1}{2}\log_2(2\pi e \sigma^2)$
     $= \frac{1}{2}\log_2 \frac{2\pi e(P+\sigma^2)}{2\pi e \sigma^2}$
     $= \frac{1}{2}\log_2(1 + \frac{P}{\sigma^2})$
  6. Holds ⇔ when $X \sim N(0, P)$

Properties of Gaussian channel cap

- depends on P & σ^2 only thru $\frac{P}{\sigma^2}$
- $= 0$ when $P = 0$
- $\frac{P}{\sigma^2} \downarrow \to C(P) \approx \frac{P}{2\sigma^2} \to \times P$, $2 \times C(P)$
- $\frac{P}{\sigma^2} \uparrow \to C(P) \approx \frac{1}{2}\log_2 \frac{P}{\sigma^2}$, $2P$, $+C$ to $C(P)$

XOR property

| self inverse: $x \oplus x = 0$

- identity: $x \oplus 0 = x$, $x \oplus 1 = \overline{x}$
- commutative: $x \oplus y = y \oplus x$ (interchangeable)
- associative: $(x \oplus y) \oplus z = x \oplus (y \oplus z)$

$u_1, \dots, u_k$ → Encoder ($\underline{x} = \underline{u}G$) → $x_1, \dots, x_n$ → channel ($\underline{y} = \underline{x} \oplus \underline{z}$) → $y_1, \dots, y_n$ → decoder → $\hat{u}_1, \dots, \hat{u}_k$

Linear Code: $m \in \{1, \dots, M\}$, $M = 2^k$ → $(u_1, \dots, u_k) \in \{0,1\}^k$

- $R = \frac{1}{n}\log_2 M \Rightarrow R = \frac{k}{n}$ ($M = 2^k$)
  max bits / source bit

Def & Gen. matrix

- systematic parity check code:
  $x_i = u_i$, $i = 1, \dots, k$
  $x_i = \bigoplus_{j=1}^{k} u_j g_{ji}$, $i = k+1, \dots, n$
  ↳ $g_{ji} = 1$ if parity check in location i includes u_j & 0 otherwise
- general ~:
  $x_i = \bigoplus_{j=1}^{k} u_j g_{ji}$, $i = 1, \dots, n$
  ↳ sys → special case → $g_{ji} := \mathbb{1}_{j=i}$ for $i = 1, \dots, k$
- $x = uG$ (mmodulo-2 arithmetic)
  ↳ sys: $G = [I_k \; P_{n-k}]$:
- u & u': $x = uG$ & $x' = u'G$
  → $x \oplus x' = uG \oplus u'G = (u \oplus u')G$
  → $(x \oplus x')$ also codeword

Parity-check (decoding)

- $xH = 0 \Leftrightarrow x$ is valid codeword
  ↳ $G = [I_k \; P] \Rightarrow H = [P \; I_{n-k}]$ (opposite order)
- Let z be noise (flips bits): $y = x \oplus z \to yH = (x \oplus z)H$
  $= xH \oplus zH$
  $= zH$

Distance

$x, x' \in \{0,1\}^n$: $d_H(x, x') = \sum_{i=1}^{n} \mathbb{1}\{x_i \neq x'_i\}$

- min distance: $d_{min} = \min_{x \neq x' \in C} d_H(x, x')$
  ↳ possible to correct up to $d_{min} - 1$ erasure & up to $\frac{d_{min}-1}{2}$ bit flips
- (claim) $d_{min} > 0 \to d_{min} = \min_{x \in C, x \neq 0} w(x)$, $w(x) = \sum_{i=1}^{n} \mathbb{1}\{x_i = 1\}$
  ↳ proof by: 1) if $d_H(x', x'') = d_{min}$, $w(x' \oplus x'') = d_{min}$ ($x \neq 0$)
  2) by linearity, $x' \oplus x''$ is also a valid codeword & $w = d_{min}$

Min Dist Decoding

- channel $P_{Y|X}$ & C, use when (ML decoder) $\hat{m} = \arg\max_{j=1,\dots,M} P_{Y|X}(y|x^{(j)})$
  ↳ (BSC) $\log P_{Y|X}(y|x) = \sum_{i=1}^{n} \log P_{Y|X}(y_i|x_i)$
  $= \sum_{i: y_i = x_i} \log(1-\delta) + \sum_{i: y_i \neq x_i} \log \delta$
  $= (n - d_H(x,y)) \log(1-\delta) + d_H(x,y) \log \delta$
  $= n \log(1-\delta) - d_H(x,y) \log \frac{1-\delta}{\delta}$ (fixed)

$\arg\max_{j=1,\dots,M} P_{Y|X} = \arg\min_{j=1,\dots,M} d_H(x^{(j)}, y)$

↳ Syndrome decoding (BSC): $\underline{s} = zH = yH$
- For linear code: $\underline{s}$: min dist codeword to $y \to \hat{z} = \arg\min w(\underline{z})$
  then $\hat{x} = y \oplus \hat{z}$ | one cos.
- (alternative look to max # errors correctable): if $\binom{n}{0} + \binom{n}{1} + \dots + \binom{n}{k}$, $k < K$ is $> 2^{n-K}$; cannot correct k bit flips (fixed)
  ↳ linear code

Other useful notes

- $d_H(x, x') = $ odd → differs odd # of pos → different parity (#1s)
  → $d_H(x, x') = w(x \oplus x') \to w(x \oplus w) = w(x) + w(x') - 2w(y \oplus x \cap x')$
  odd → odd fewer | even → even "o" even
- parity check bit: 1 if odd #1s 0 otherwise
  ↳ $c_1 + c_2 + \dots + c_n + P = 0 \pmod 2 \to P = \bigoplus c_i$
- non-sys → sys: ① swap two rows ② $g_i \oplus g_j = g_i$ ③ Gaussian elimination
- Σ of cont & discrete RV: ($Z = X + Y$) $f_Z(z) = ?$
  ↳ $F_Z(z) = \mathbb{P}[Z \le z] = \sum_x \mathbb{P}[X+Y \le z | X = x] P_X(x) = \dots = \sum_x F_Y(z-x) P_X(x)$
  ↳ $f_Z(z) = \frac{d}{dz} F_Z(z) = \sum_x \frac{d}{dz} F_Y(z-x) P_X(x) = \sum_x f_Y(z-x) P_X(x)$
- cond ⊥ → ⊥ | $C \le \min\{\log_2|\mathcal{X}|, \log_2|\mathcal{Y}|\}$
- $P = [p_1, p_2, \dots, p_n]$: $H(P) = H(p_1, 1-p_1) + (1-p_1) H(\frac{p_2}{1-p_1}, \dots, \frac{p_n}{1-p_1})$
  (decomposability of Entropy)
- $I(X;U|V) = \sum_v P_V(v) I(X;U|V=v)$

Scanned with CamScanner